# CSE/ISyE 6740
# Computational Data Analysis

## Fall 2026

Kai Wang, Assistant Professor in Computational Science and Engineering
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Based on materials by Anqi Wu, B. Aditya Prakash, Le Song, Mahdi Roozbahani, Carlos Guestrin

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# Course Information

- CSE/ISyE 6740 Computational Data Analysis
  - **Date and time:** Monday and Wednesday, 12:30pm – 1:45pm
  - **Classroom**: TSRB Room 132
  - **Format**: in-class lectures, no recording

- Instructor: Kai Wang
  - Assistant Professor, CSE, College of Computing
  - **Office**: CODA S1309
  - **Email**: kwang692@gatech.edu
  - **Research Interests**: AI for social impact, machine learning, optimization, reinforcement learning, online learning, multi-agent systems, diffusion models

  - **Office hours**: Monday 3-4pm ET KACB 3121 (tentatively)

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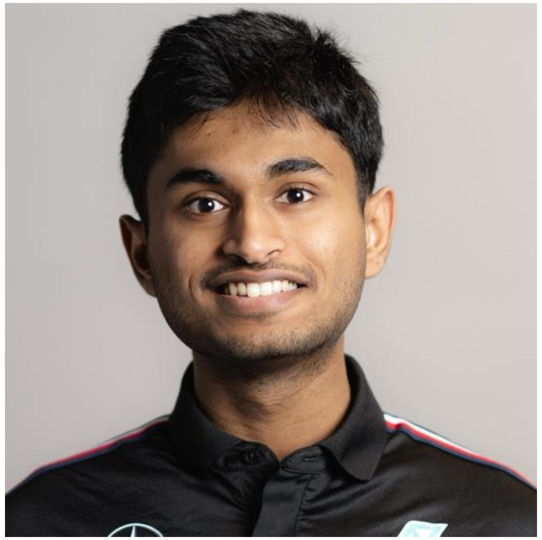
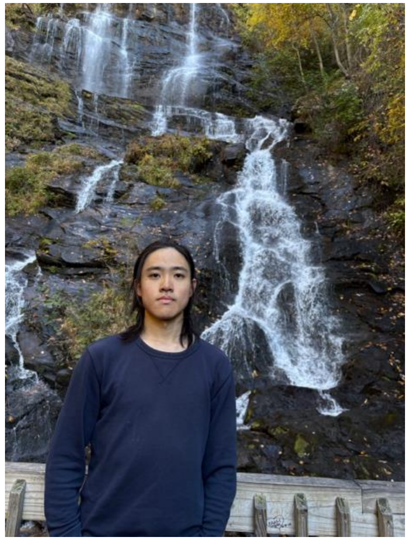
# Teaching Assistants

Shreeyash Gowaikar (head TA)
sgowaikar3@gatech.edu
Wednesday 2:30 - 3:30pm
KACB 2nd Floor



Soumil Sahu
ssahu64@gatech.edu
Tuesday 10 -11am
KACB 1st floor open space



Adrian Ho
sho73@gatech.edu
Friday 2 - 3pm
CODA 2nd floor

They will hold office hours and answer questions on Piazza and Gradescope for you. Both have been activated on Canvas.

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# Piazza and Gradescope

- This term, we will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the TA, and myself. Rather than emailing questions to the teaching staff, I encourage you to post your questions on Piazza
- Piazza link: https://piazza.com/gatech/fall2026/cse6740a/home
- Gradescope link: https://www.gradescope.com/courses/1338210
- You can also find the Piazza link on your Canvas page.

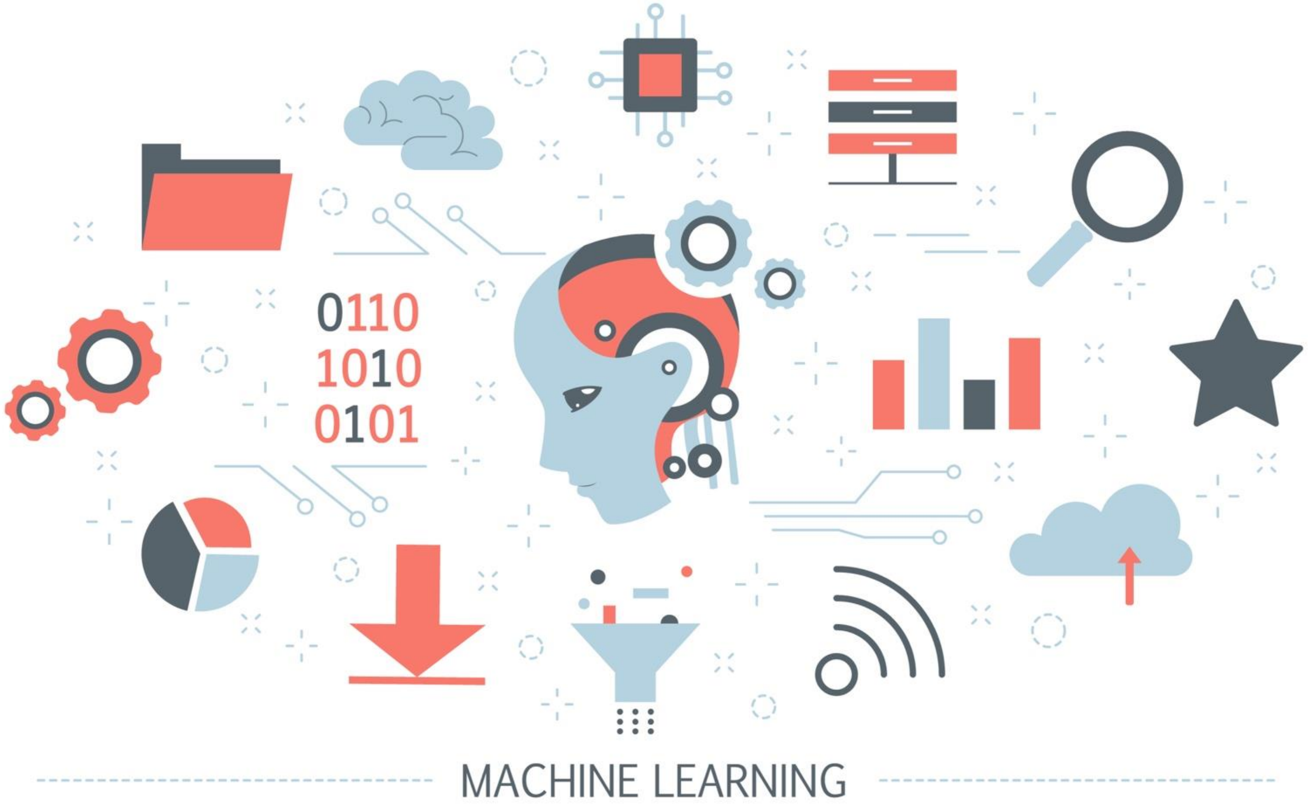
# Outline

- Instructor, TAs, and course information
- Course overview
- Basics and prerequisites (with linear regression as an example)
- Assignment and grading
- Homework 0

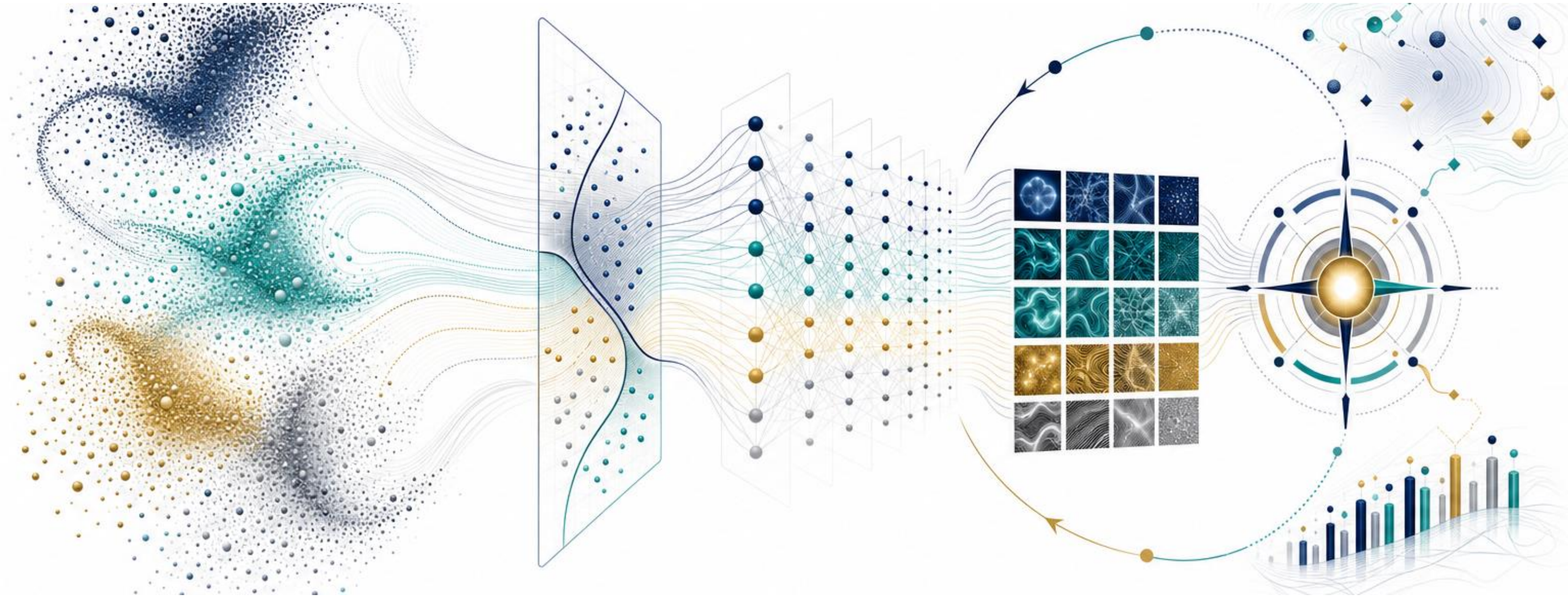
# Course Overview

# What is Machine Learning (ML)?

- Study of algorithms that improve their performance at some tasks based on experience



# Syllabus



## Unsupervised learning
What patterns are already present in unlabeled data?

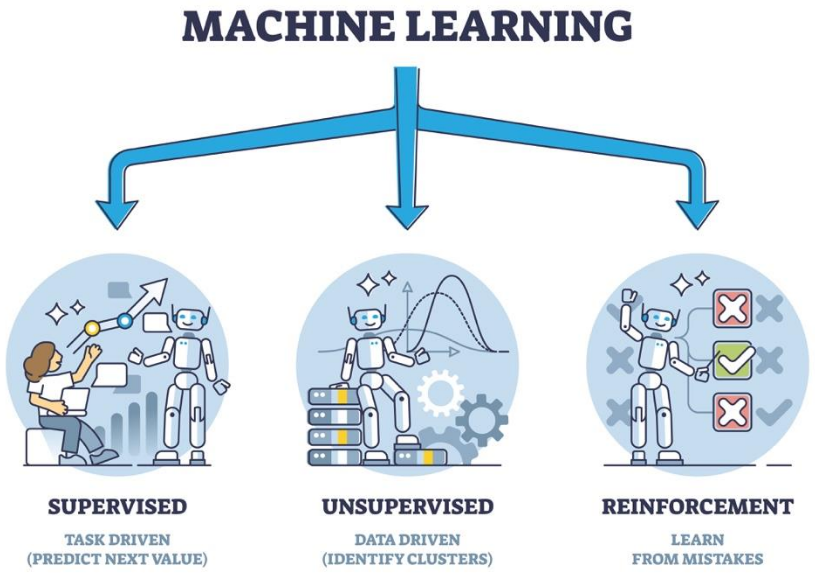
## Supervised learning
How do features map to labels or numerical targets?

## Neural networks
What useful abstractions can deep models construct?

## Reinforcement learning
How should an agent act when choices affect the future?

# Syllabus

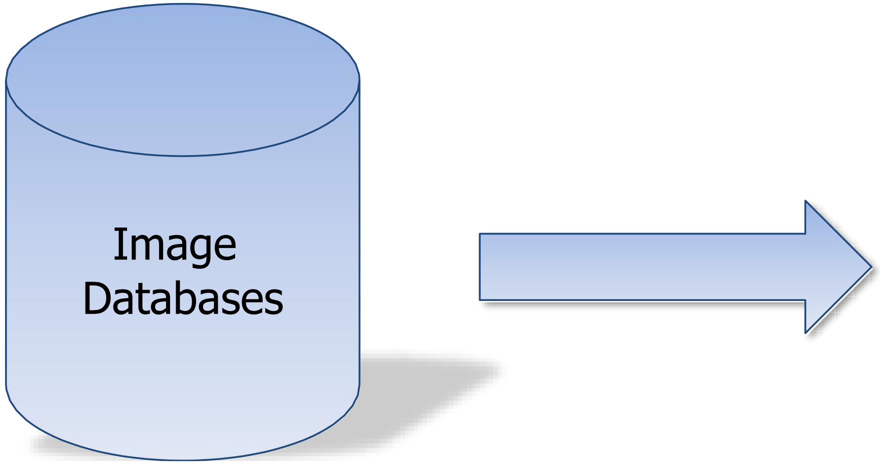
- Machine learning turns data and experience into models that discover, predict, generate, and decide.
- **Course journey**
  - **Unsupervised learning | Week 1-3**
    - Clustering, PCA, density estimation, Gaussian mixtures, EM
  - **Supervised learning | Week 4-7**
    - Trees, logistic regression, SVM, kernels/GPs, neural networks, optimization
  - **Neural networks | Week 9-11**
    - CNNs, autoencoders/VAEs, diffusion, RNNs, transformers
  - **Reinforcement learning | Week 12-14**
    - Planning, model-free reinforcement learning, deep RL

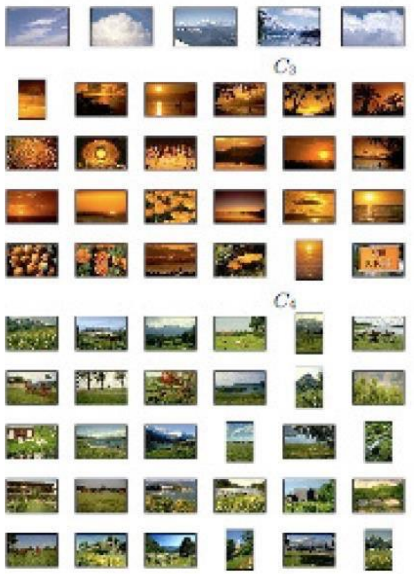


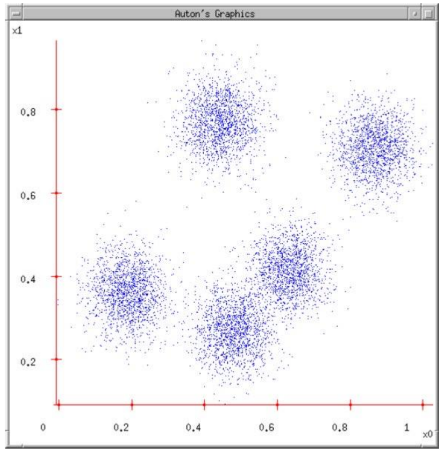
# Syllabus: Unsupervised Learning

- Learning without labels or without optimizing for prediction tasks
  - **Clustering vectorized data (week 1)**
    - K-means clustering
    - Hierarchical clustering
    - Graph clustering
  - **Dimensionality reduction (week 2)**
    - Principal component analysis
    - Nonlinear dimensionality reduction
  - **Density estimation (week 3)**
    - Feature selection
    - Novelty/abnormality detection
    - Gaussian mixture models, EM algorithm

# Organizing Images

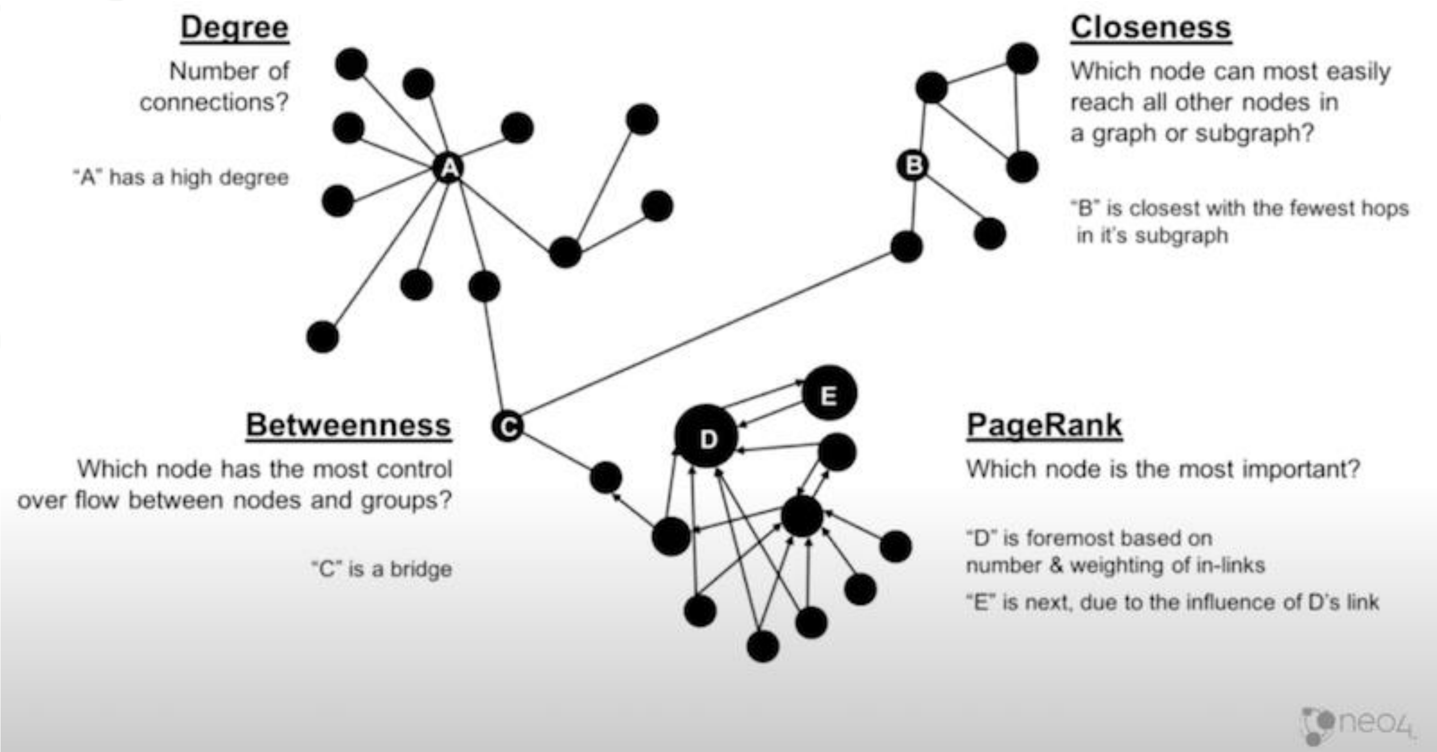





- What are the desired outcome?
- What are the input (data)?
- What are the learning paradigms?

# Find Communities in Social Network




- What are the desired outcome?
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# Visualize Image Relations


- What are the desired outcome?
- What are the input (data)?
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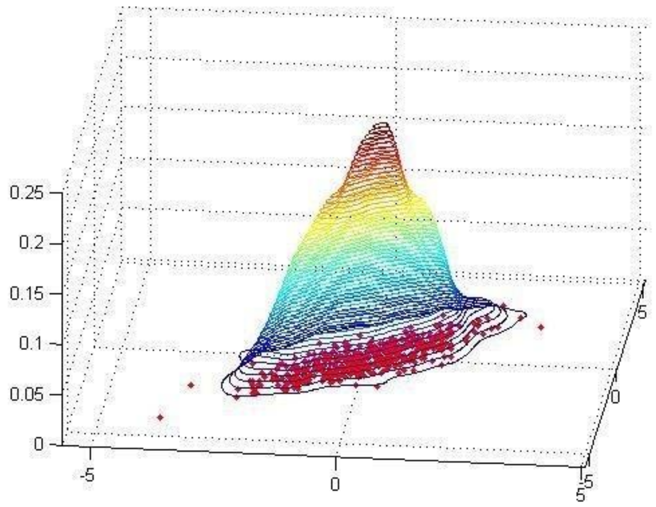
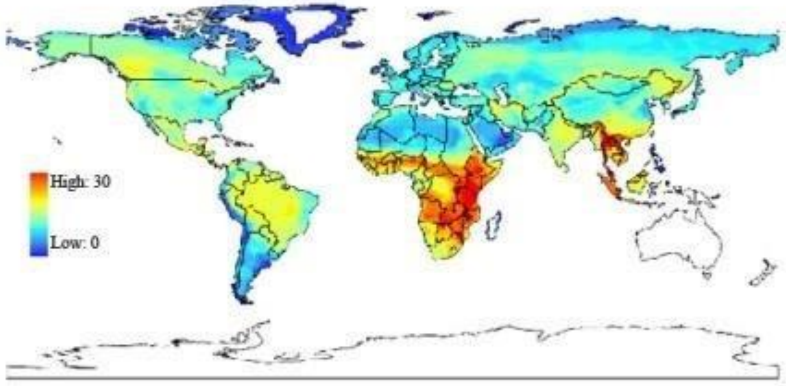
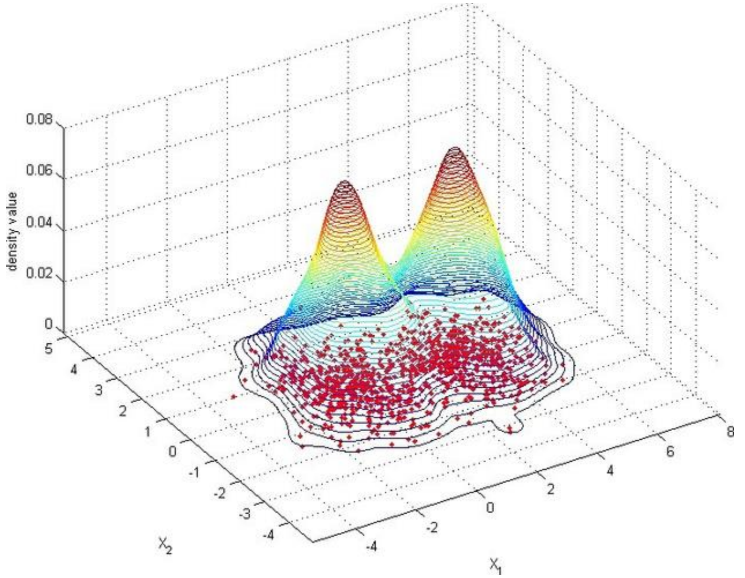
# Shape of Data




Number of carnivore species found in different regions across the globe




- What are the desired outcome?
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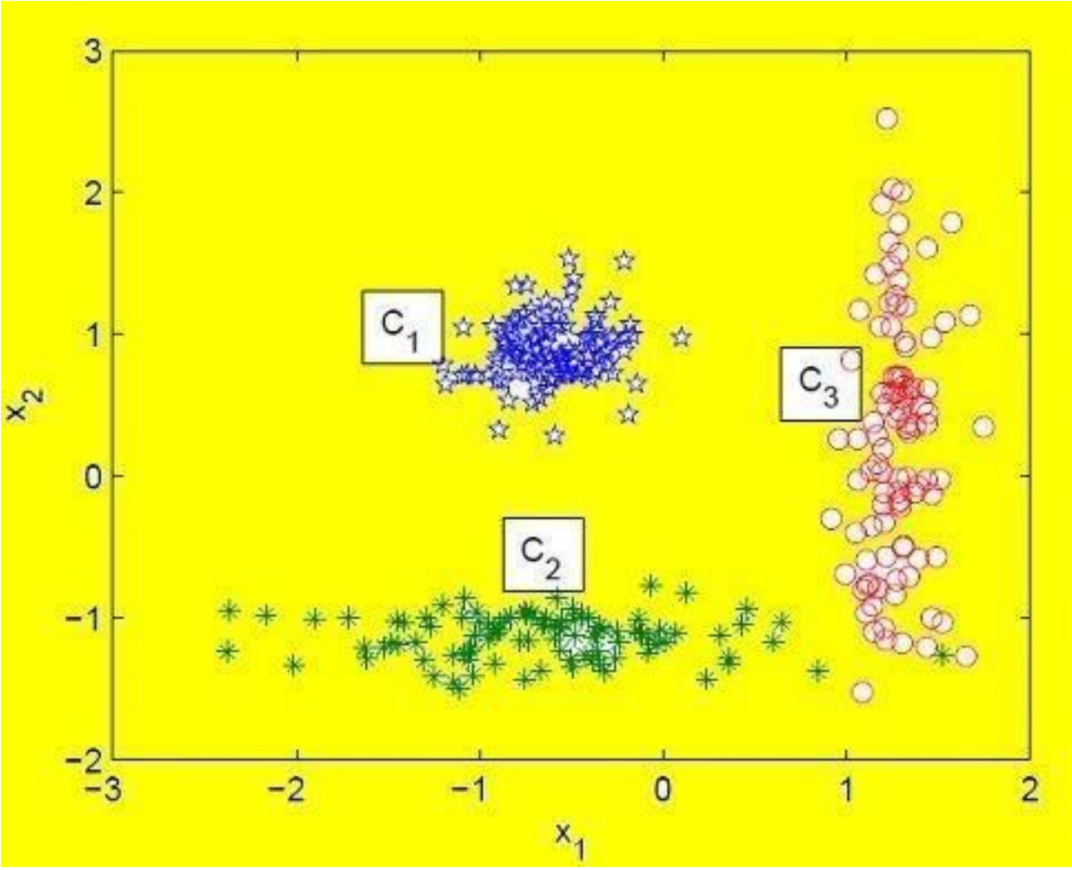
# Feature Selection





- What are the desired outcome?
- What are the input (data)?
- What are the learning paradigms?

# Novelty/anomaly Detection

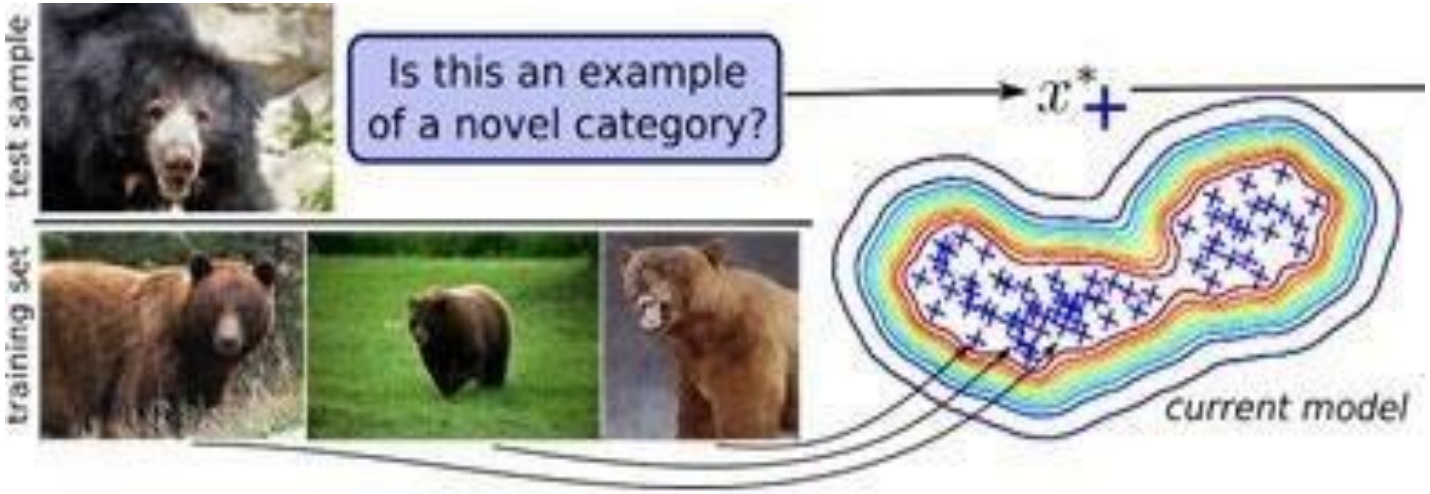


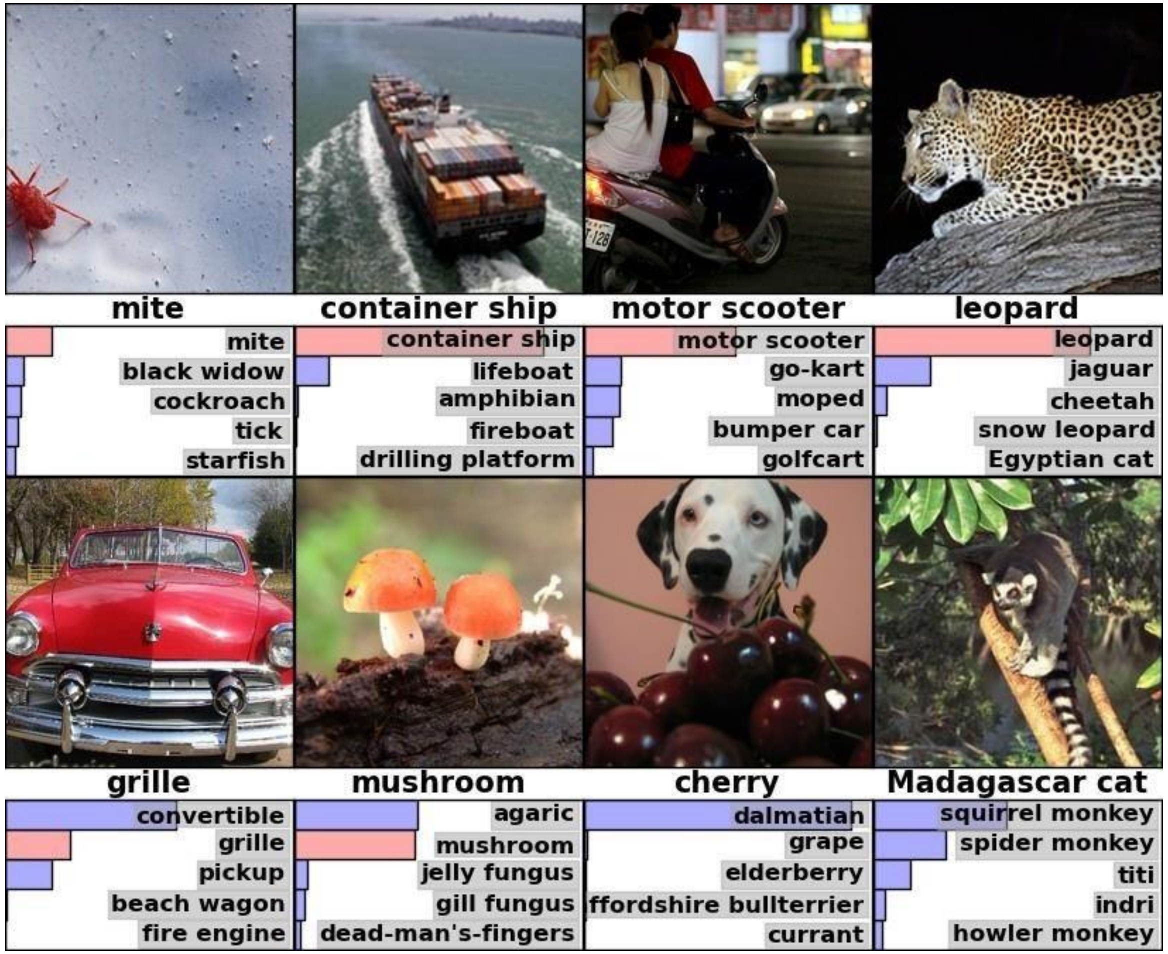

FIGURE 1 EXAMPLE OF NOVELTY DETECTION, BODESHEIM (2012).

- What are the desired outcome?
- What are the input (data)?
- What are the learning paradigms?

# Syllabus: Supervised Learning

- Learning with labels, focusing on predictive performance
  - **Classical classification models (week 4)**
    - K-nearest neighbors, Naive bayes, logistic regression, decision trees
  - **Support vector machine (week 5)**
    - Convex analysis, primal and dual, SVM
  - **Kernel methods (week 6)**
    - Kernel regression, Bayesian regression, Gaussian process regression
  - **Neural networks (week 7)**
    - Gradient descent, neural networks, overfitting, bias-variance tradeoff

# Image Classification




- What are the desired outcome?
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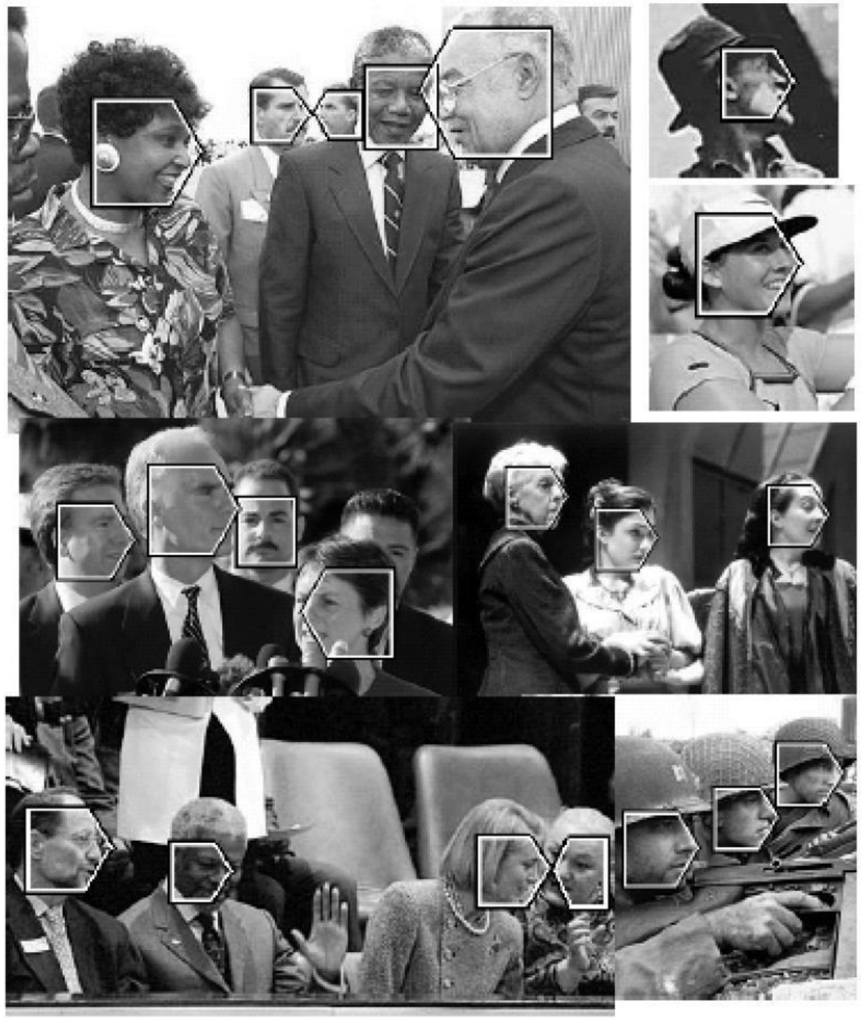
# Face Detection



- What are the desired outcome?
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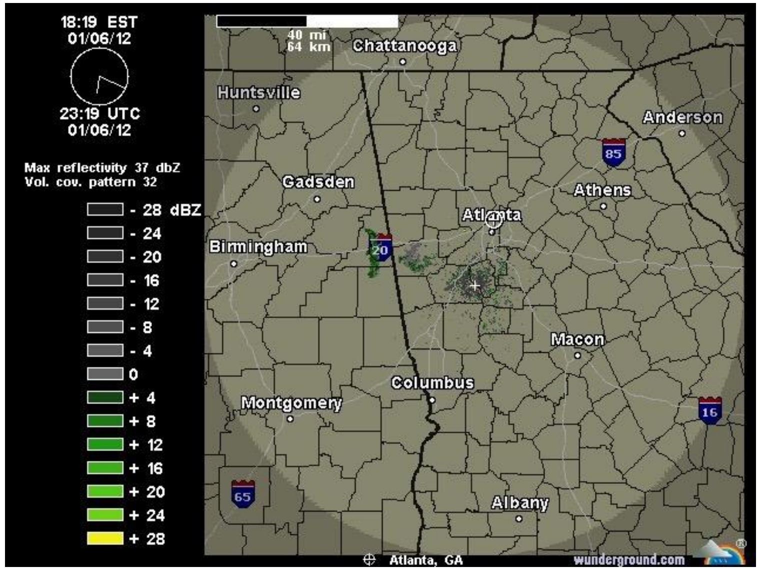
# Weather Prediction



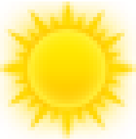

Predict

Numeric values: 40F

Wind: NE at 14km/h

Humidity: 83%

Predict



- What are the desired outcome?
- What are the input (data)?
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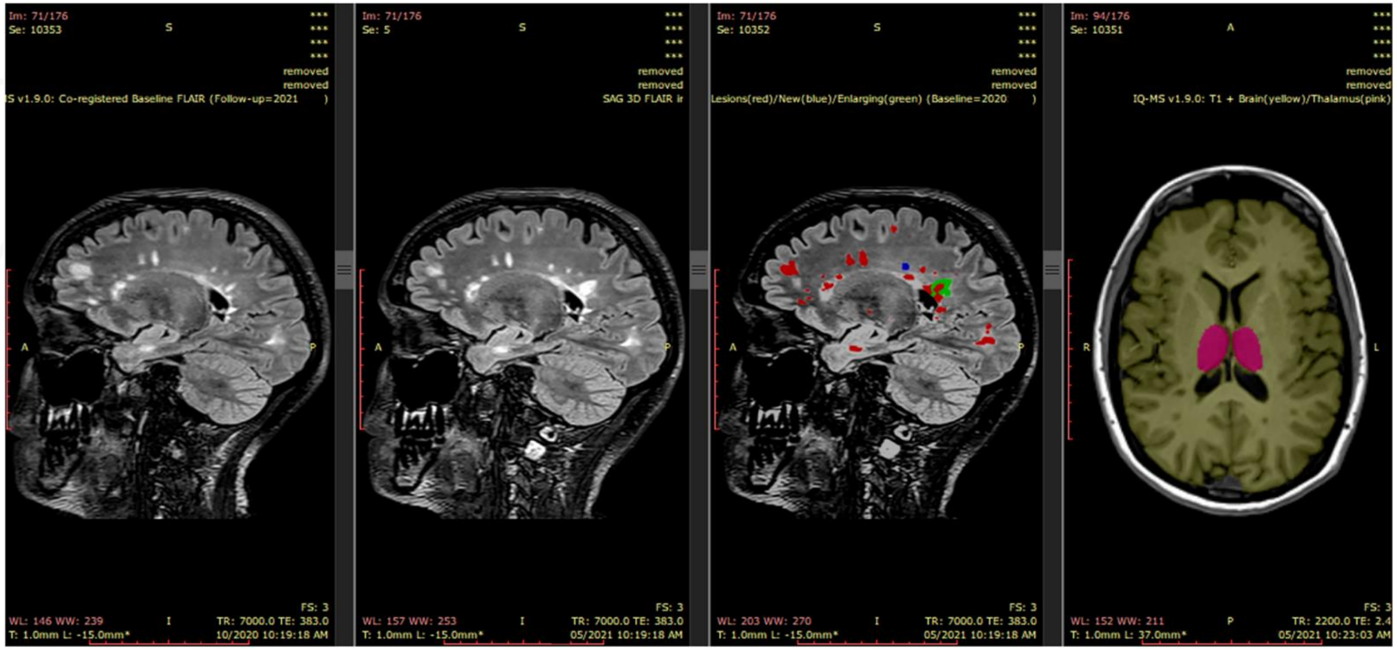
# Understanding Brain Activity

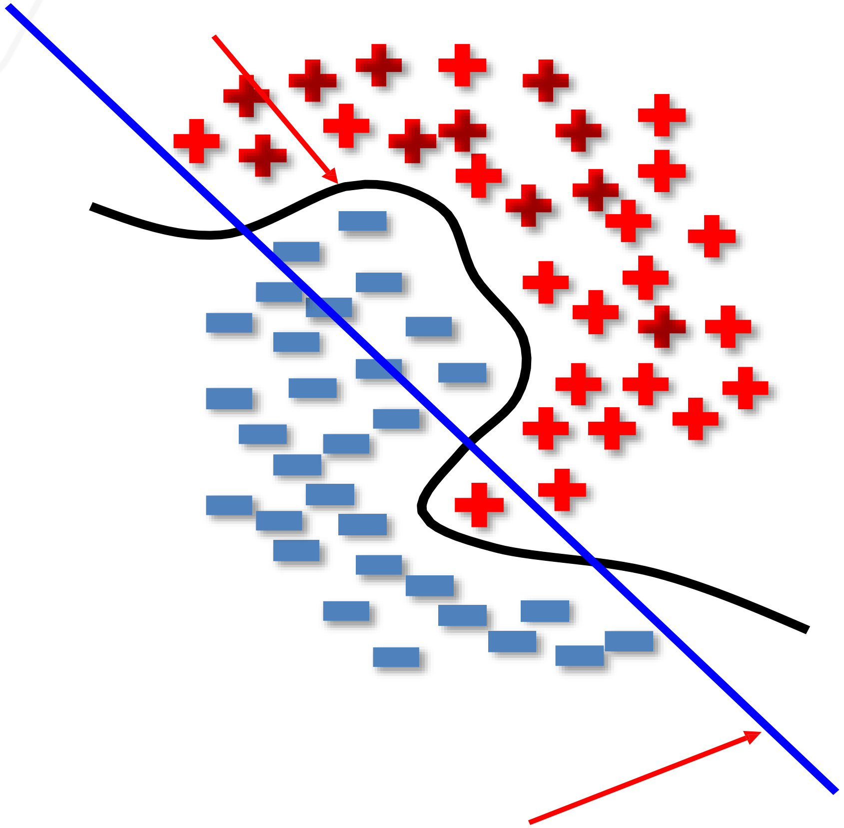



“A real-world clinical validation for AI-based MRI monitoring in multiple sclerosis”
Barnett et al., Nature 2023

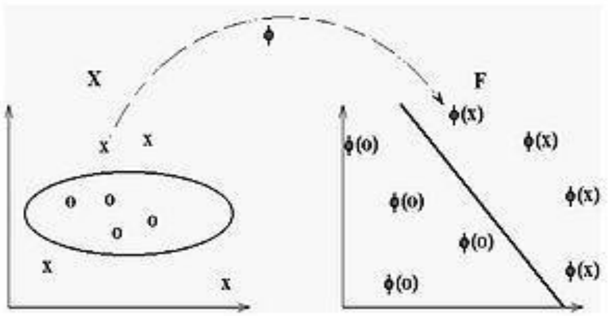
- What are the desired outcome?
- What are the input (data)?
- What are the learning paradigms?

# Non-linear Classifier

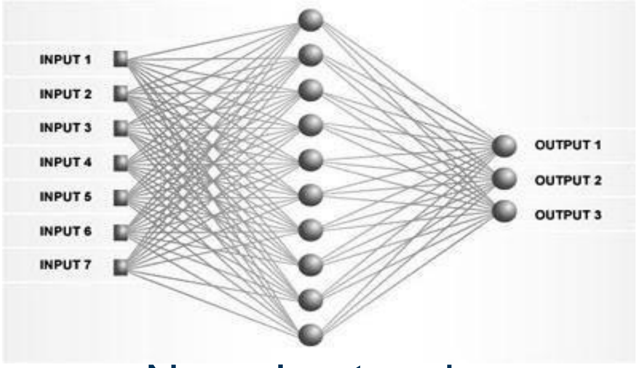
Non-linear decision boundary



Linear SVM decision boundary




Kernel methods




Neural networks

- What are the desired outcome?
- What are the input (data)?
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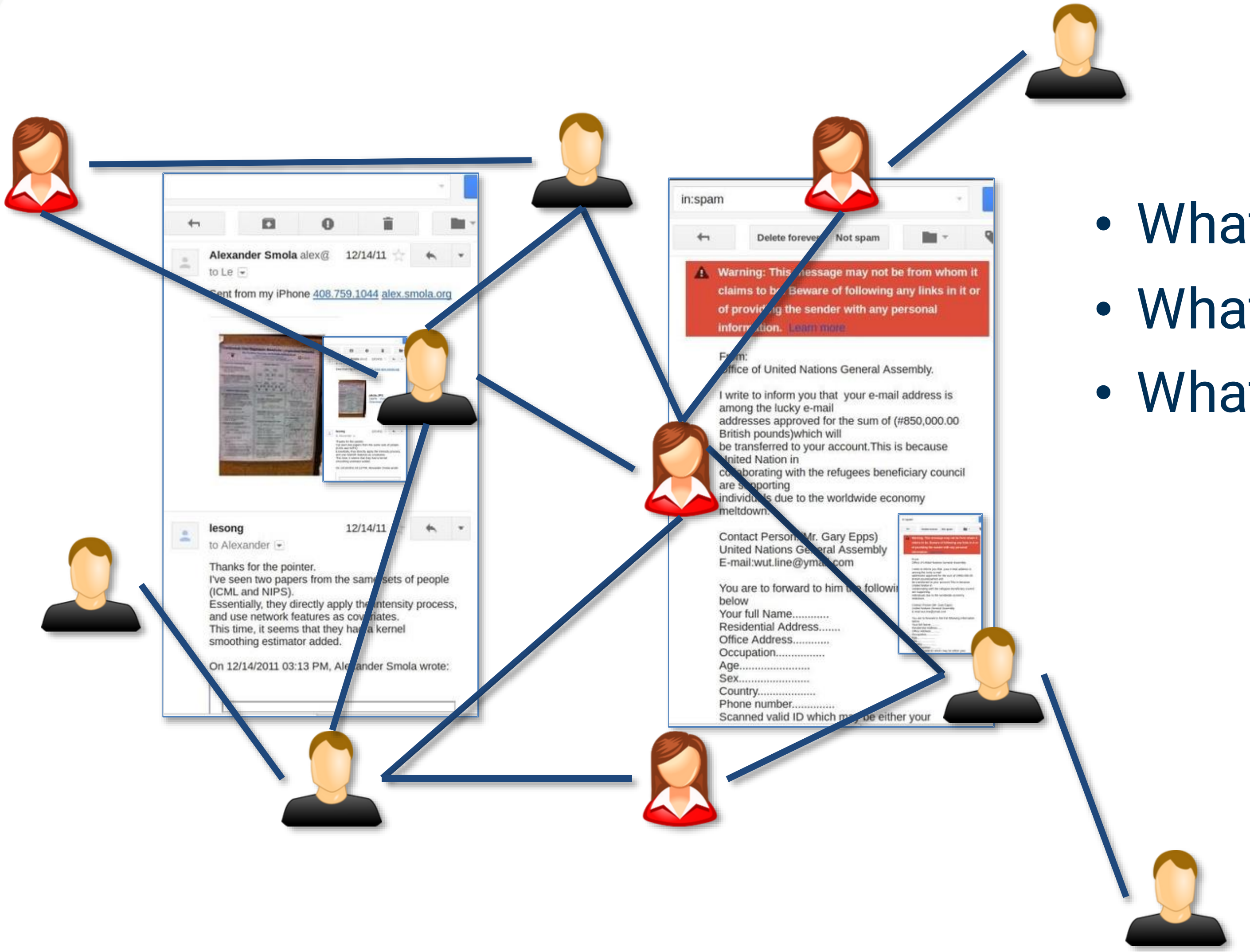
# Spam Filtering




- What are the desired outcome?
- What are the input (data)?
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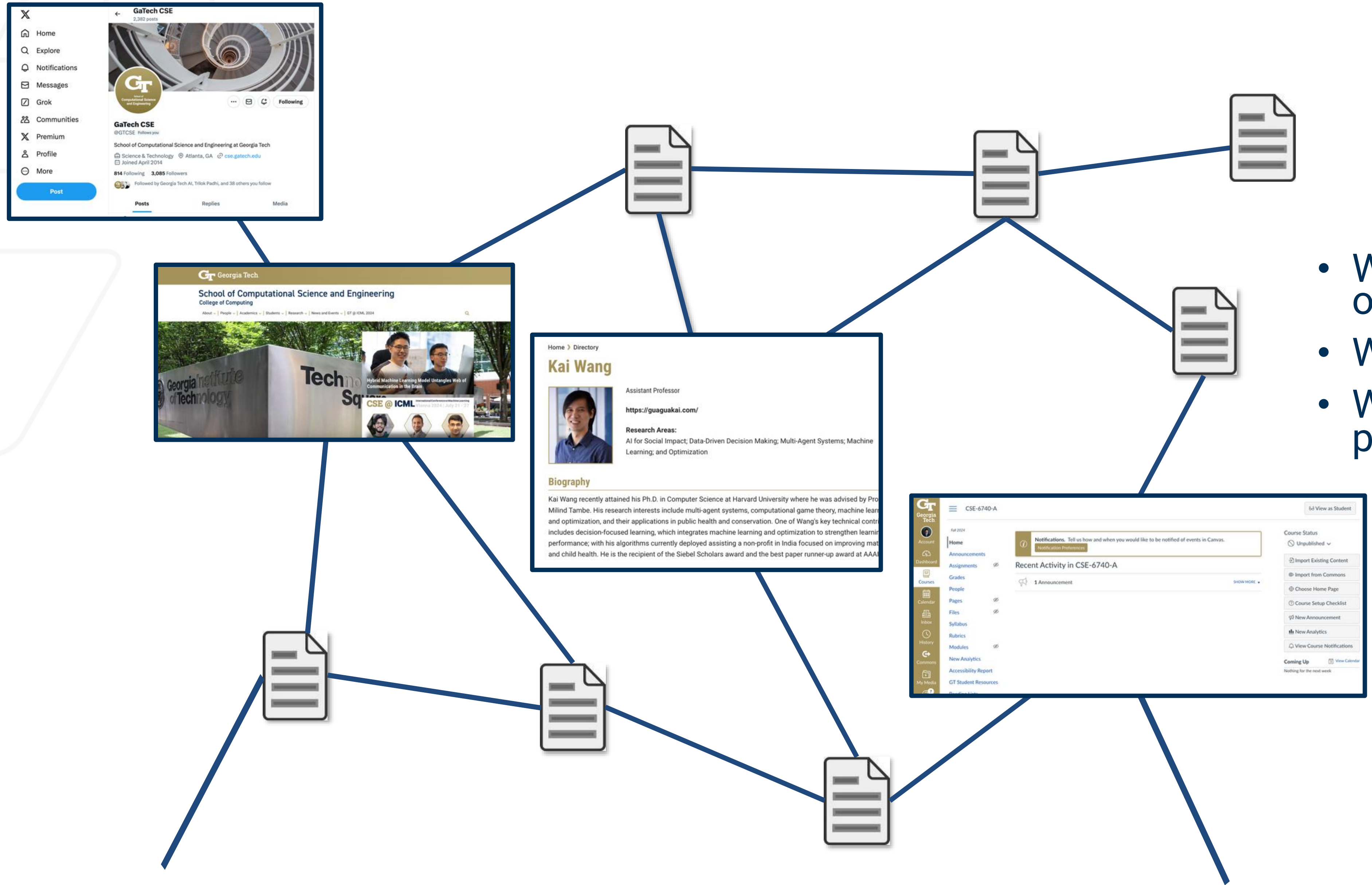
# Webpage Classification




- What are the desired outcome?
- What are the input (data)?
- What are the learning paradigms?

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# Syllabus: Advanced Topics

- **Neural networks (week 9-11)**
  - Convolutional neural networks
  - Deep learning
  - Autoencoders, variational autoencoders
  - Diffusion models

- **Reinforcement learning (week 12-14)**
  - Dynamic programming
  - Bellman equation
  - Q-learning and SARSA
  - Deep RL

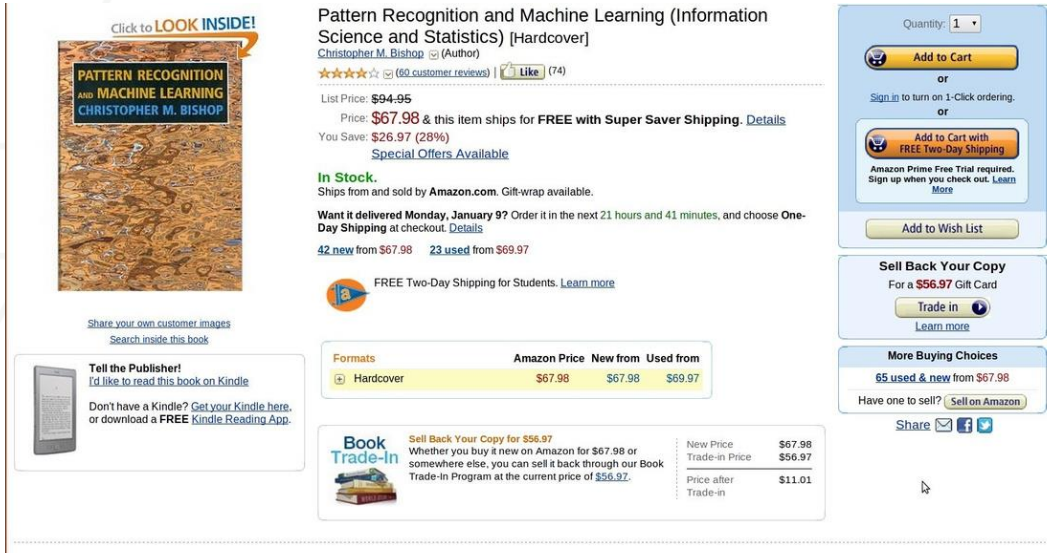
# Handwritten Digit Recognition / Text Annotation

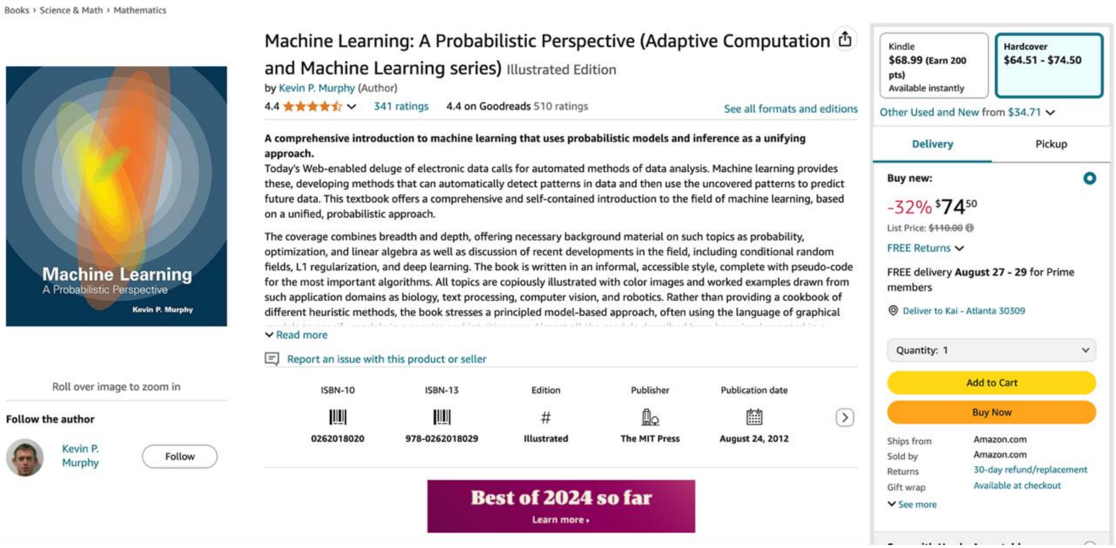
Inter-character dependency

Inter-word dependency

Aoccdrnig to a sudty at Cmabrigde Uinervtisy, it deosn't mttaer in waht oredr the ltteers in a wrod are, the olny iprmoetnt tihng is taht the frist and lsat ltteer be at the rghit pclae. The rset can be a ttoal mses and you can sitll raed it wouthit aporbelm. Tihs is bcuseae the huamn mnid deos not raed ervey lteter by istlef, but the wrod as awlohe.

# Product Recommendation





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# Recommendation with Human Feedback


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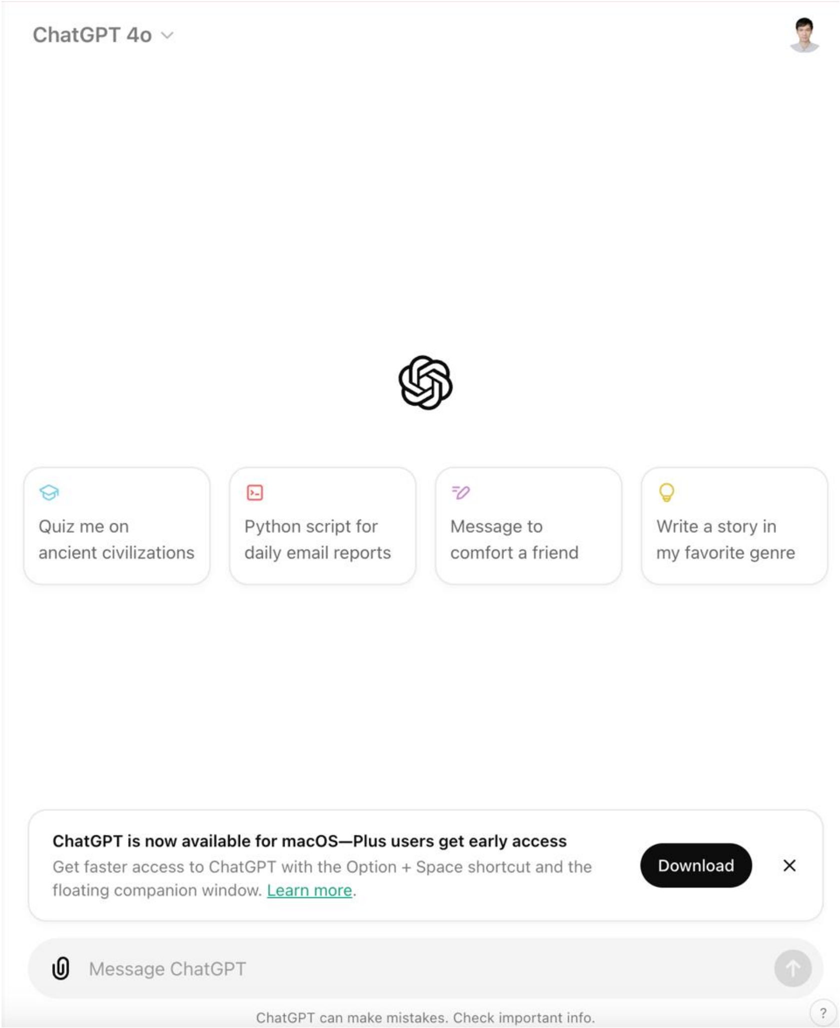
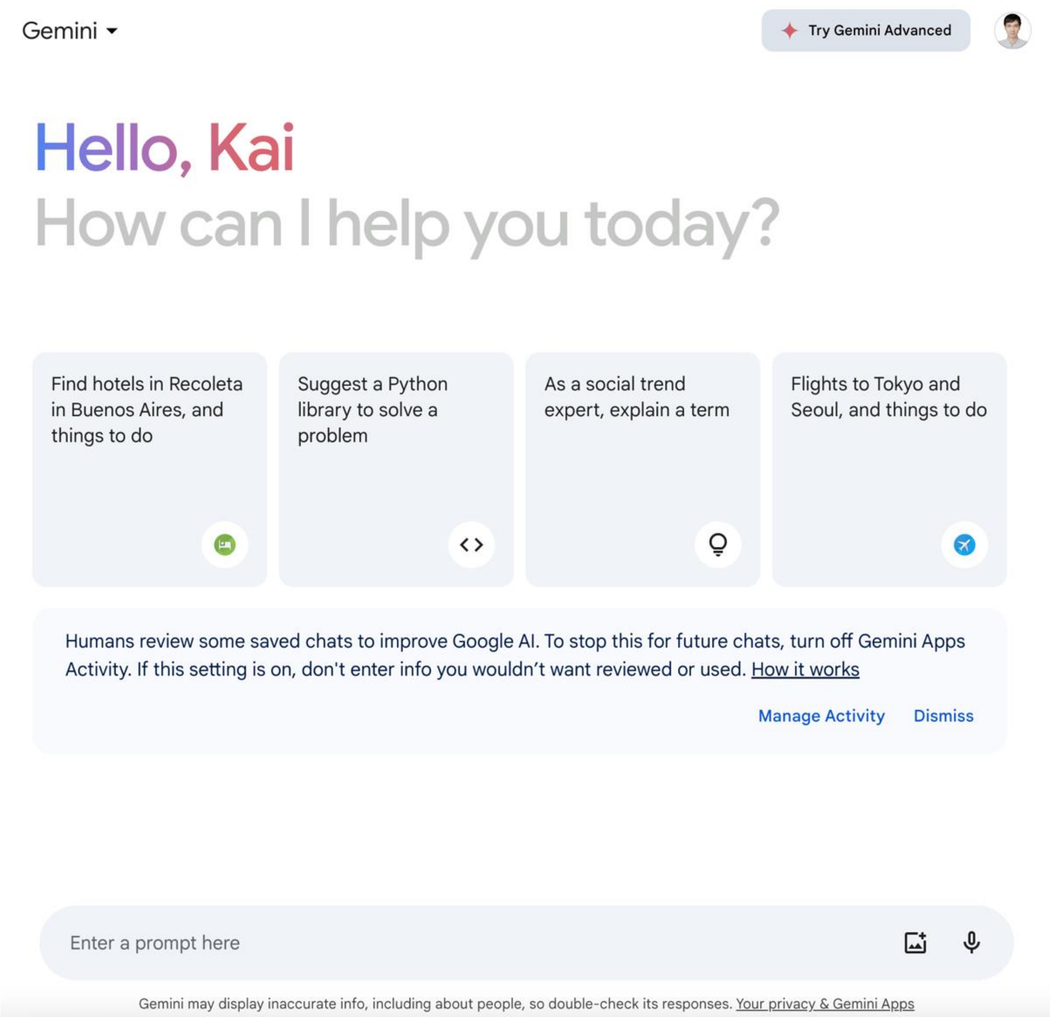
# Language Models



# Image Generation

# AlphaGo


柯洁 KE JIE
00:13:28
ALPHAGO
01:30:35
Google
中国乌镇 围棋峰会
Future of Go Summit in Wuzhen

# Robotics


# Basics / Prerequisites

# Basics / Prerequisites

- **Probability**
  - Distributions, densities, marginalization, conditioning
- **Statistics**
  - Mean, variance, maximum likelihood estimation
- **Linear algebra**
  - Vector, matrix, multiplication, inversion, eigen-value decomposition
- **Algorithm, programming, and optimization**

# Homework 0

- We highly recommend that you check out **Homework 0** as soon as possible to see if you feel comfortable with the prerequisites!!

- Homework 0 is graded by SAT/UNSAT. As long as you submit in time, you get full credit. But please use this opportunity to reassess if you meet all the prerequisites of CSE6740.

- The amount of time spent on each homework (expected): **20+ hours**
  - This varies a lot depending on your background. It can easily go over 40+ hours if you don't have the right prerequisites, and we also don't want you to feel overwhelmed by homework.

| Question Text | N | RR | Interpol. Median | 3 | 6 | 9 | 12 | 15 | 18 | 18+ |
|---|---|---|---|---|---|---|---|---|---|---|
| Hours per week spent on course | 63 | 53% | 14.25 | 0 | 2 | 7 | 15 | 10 | 14 | 15 |

# Textbooks

- "Pattern Recognition and Machine Learning", Christopher M. Bishop
- "The Elements of Statistical Learning", Trevor Hastie, Robert Tibshirani, Jerome Friedman

# Linear Regression (Prerequisite Example)

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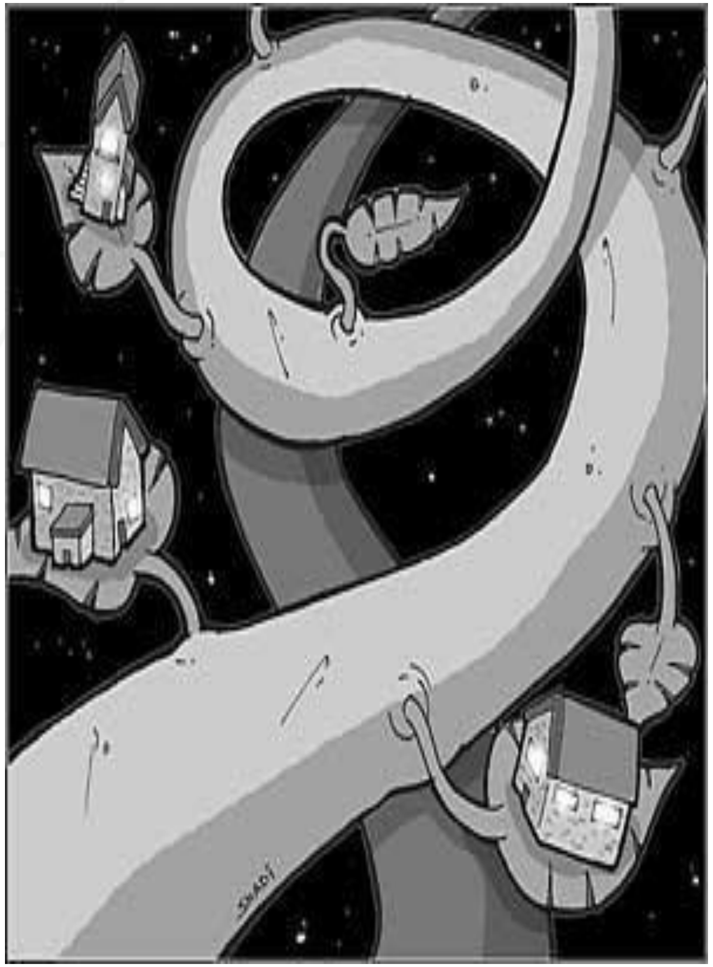
# Machine Learning for Apartment Hunting



- Suppose you want to move to Atlanta, and you want to find the most **reasonably priced** apartment satisfying your **needs** (I know it is hard and probably too late):

| Living area ($\mathrm{ft}^2$) | # bedroom | Monthly rent ($) |
|---|---|---|
| 230 | 1 | 900 |
| 506 | 2 | 1800 |
| 433 | 2 | 1500 |
| 190 | 1 | 800 |
| … | | |
| 150 | 1 | ? |
| 270 | 1.5 | ? |

# Linear Regression Model

- Assume $y$ is a linear function of $x$ (features) plus noise $\epsilon$

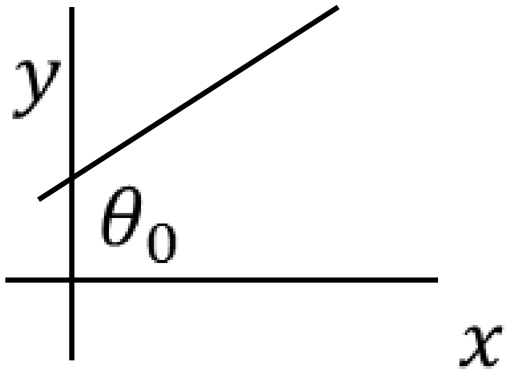
$$y = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d + \epsilon$$

where $\epsilon$ is an error modeled as Gaussian $N(0, \sigma^2)$ Probability

- Let $\theta = (\theta_0, \theta_1, \dots, \theta_d)^\top \in \mathbb{R}^{d+1}$, and augment data by one dimension

Linear algebra

$$x \leftarrow (1, x)^\top$$




Then $y = \theta^\top x + \epsilon$

Linear algebra

# Ordinary Least Squares Method

- Given $n$ data points, find $\theta$ that minimizes the mean squared error

$$\hat{\theta} = \mathrm{argmin}_{\theta} L(\theta) := \frac{1}{n}\sum_{i=1}^{n}\left(y^i - \theta^\top x^i\right)^2$$

Optimization

Statistics

- Set gradient to 0 to find the optimal parameter:

Calculus

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{n}\sum_{i=1}^{n}\left(y^i - \theta^\top x^i\right) x^i = 0$$

$$\Leftrightarrow -\frac{2}{n}\sum_{i=1}^{n} y^i x^i + \frac{2}{n}\sum_{i=1}^{n} x^i {x^i}^\top \theta = 0$$

Linear algebra

# Matrix Version of the Gradient

- Define $X = (x^1, x^2, \dots, x^n)^\top \in \mathbb{R}^{n \times (d+1)}$, $y = (y^1, y^2, \dots, y^n)^\top \in \mathbb{R}^n$, the gradient becomes

Linear algebra

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{n} X^\top y + \frac{2}{n} X^\top X \theta = 0$$

Linear algebra

$$\Rightarrow \hat{\theta} = (X^\top X)^{-1} X^\top y$$

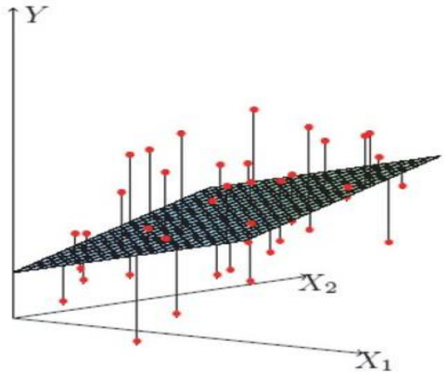
Algorithms
Programming

- Matrix inversion in $\hat{\theta} = (X^\top X)^{-1} X^\top y$ is often **expensive** to compute.
  - Alternatively, we can use gradient descent:

$$\hat{\theta}^{t+1} \leftarrow \hat{\theta}^t + \frac{\alpha}{n} \sum_{i=1}^{n} \left( y^i - \hat{\theta}^{t\top} x^i \right) x^i$$

Optimization

# Probabilistic Interpretation

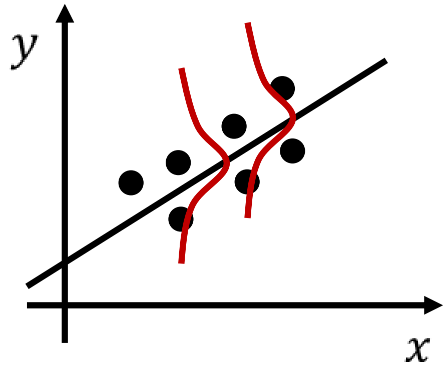



- Assume $y$ is linear in $x$ plus noise $\epsilon$

$$y = \theta^\top x + \epsilon$$

- Assume $\epsilon$ follows a Gaussian $N(0, \sigma^2)$

$$p(y^i | x^i; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\left(y^i - \theta^\top x^i\right)^2}{2\sigma^2}\right)$$




- By independence assumption, likelihood is:

$$L(\theta) = \prod_{i=1}^{n} p(y^i | x^i; \theta) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left(-\frac{\sum_{i=1}^{n}\left(y^i - \theta^\top x^i\right)^2}{2\sigma^2}\right)$$

Probability

# Probabilistic Interpretation

- Hence, the log-likelihood can be written as:

$$MLE:\ \log L(\theta) = n \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(y^i - \theta^\top x^i\right)^2$$

Statistics

- Therefore, OLS is equivalent to MLE of $\theta$

$$LMS:\ \frac{1}{n}\sum_{i=1}^{n}\left(y^i - \theta^\top x^i\right)^2$$

- How to make it work in real data?

Algorithms
Programming

# Assignment and Grading

# Assignment

- There will be four homework (hw1 – hw4) + one homework letting you familiarize yourself with Gradescope (hw0).
- Each homework will come with both a written part and a programming part.
- **Deadline**: Two weeks after the release date (Sunday 11:59pm).
- **Coding language**: Python

- **Collaboration policy**: we think peer discussion is the key to learn and grow, and we strongly recommend discussing the course materials and homework with your peers. Please **mention your collaborator(s) (and on what topics)** in the beginning of your homework when submitting it.

  However, all work you submit must be your own. You should never include in your assignment anything that was not written directly by you without proper citation (including quotation marks and LLM generated texts).

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# Academic Integrity

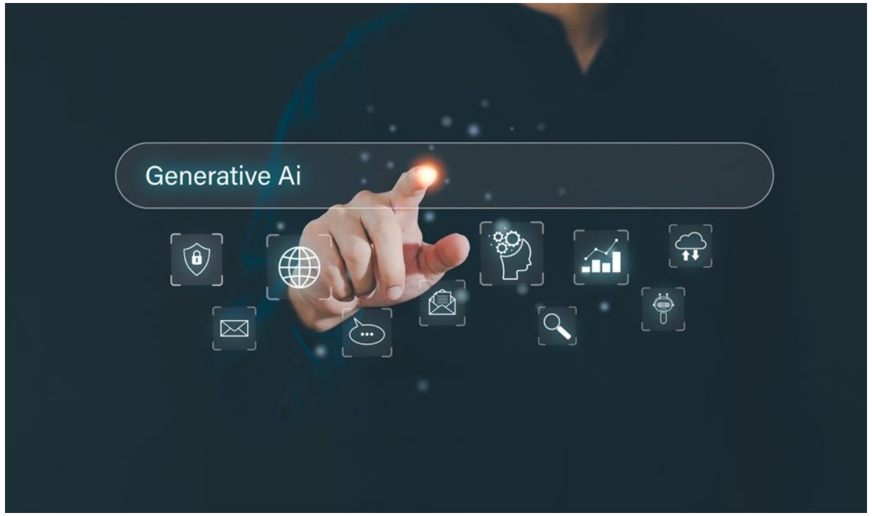
- Any evidence of cheating or other violations will be referred to the Dean of Students with a recommendation that the penalty be an award of zero points for the graded requirement, and a one letter grade reduction in the course.

- Cheating includes, but is not limited to:
  - Using unauthorized references or notes
  - Copying directly from any source, including friends, classmates, tutors, or a solutions manual
  - Allowing another person to copy your work
  - Taking an exam or handing in a graded requirement in someone else's name, or having someone else take an exam or hand in a graded requirement in your name; or asking for a re-grade of a paper that has been altered from its original form.

- Any student suspected of cheating or plagiarizing on a quiz, exam, or assignment will be reported to the Office of Student Integrity, who will investigate the incident and identify the appropriate penalty for violations.

# The Use of Generative AI

- This course is about growing in your ability to write, communicate, and think critically. Generative AI agents such as ChatGPT, DALL-E 2, and others present great opportunities for learning and for communicating.
- <u>However, AI cannot learn or communicate for you, and so cannot meet the course requirements for you</u>.
- In this course, using generative AI tools in the work of the course (including assignments, discussions, ungraded work, etc.) is allowed only in instances specified by your instructor.

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# What Do You Want to Get From This Course?

If you are interested in **study group**, please come to the front after the course and Shreeyash can collect your information and arrange accordingly.

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# Gradescope: Homework Submission

- Please use Gradescope to submit your homework.
  - You should submit both answers to the written and the programming questions (if any) in **one single PDF**.
  - You should also **submit your programming solution** in a single Jupyter notebook file.
  - We will use autograder to grade most of the programming questions. You should be able to see your scores on the public test cases.

- Homework late submission penalty
  - **20% penalty per late day**. Maximum 5 late days.
  - So please start working on your homework earlier!!

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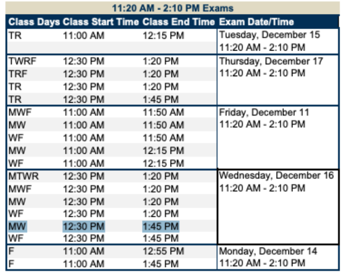
# Grading

| Assignment | Weight (Percentage, points, etc) | Date |
|---|---|---|
| Homework | 45% in total<br>• 5% for hw0 (graded SAT/UNSAT)<br>• 10% for each hw1-hw4 | See course schedule on Canvas |
| Midterm (in person) | 20% | 10/14/2026 |
| Final (in person) | 35% | 12/16/2026, 11:20AM – 2:10PM |



11:20 AM - 2:10 PM Exams

| Class Days | Class Start Time | Class End Time | Exam Date/Time |
|---|---|---|---|
| TR | 11:00 AM | 12:15 PM | Tuesday, December 15 11:20 AM - 2:10 PM |
| TWRF | 12:30 PM | 1:20 PM | Thursday, December 17 11:20 AM - 2:10 PM |
| TRF | 12:30 PM | 1:20 PM | |
| TR | 12:30 PM | 1:20 PM | |
| TR | 12:30 PM | 1:45 PM | |
| MWF | 11:00 AM | 11:50 AM | Friday, December 11 11:20 AM - 2:10 PM |
| MW | 11:00 AM | 11:50 AM | |
| WF | 11:00 AM | 11:50 AM | |
| MW | 11:00 AM | 12:15 PM | |
| WF | 11:00 AM | 12:15 PM | |
| MTWR | 12:30 PM | 1:20 PM | Wednesday, December 16 11:20 AM - 2:10 PM |
| MWF | 12:30 PM | 1:20 PM | |
| MW | 12:30 PM | 1:20 PM | |
| WF | 12:30 PM | 1:20 PM | |
| MW | 12:30 PM | 1:45 PM | |
| WF | 12:30 PM | 1:45 PM | |
| F | 11:00 AM | 12:55 PM | Monday, December 14 11:20 AM - 2:10 PM |
| F | 11:00 AM | 1:45 PM | |

| Extra credit opportunity | Weight (Percentage, points, etc) | Date |
|---|---|---|
| **LaTeX class notes (individual)** | 3% (graded SAT/UNSAT) – first come first serve, notes in latex | Voluntary, starting from Week 2 |
| **Advanced homework questions** | There will be additional challenging questions in the homework that you can solve and get bonus. | To be assigned in each homework |

Please see the announcement on Piazza and Canvas for signing up for the note taking.

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# Grading

- **Homework**: 45%
  - Homework 0:
    - 5%, graded SAT/UNSAT.
    - Familiarizing yourself with how to use Gradescope and some background knowledge!
  - Homework 1-4:
    - 10% each, due two weeks after the homework is released (no later than the end of Sunday). Due time is on **Sunday 11:59pm** (unless further notice).
- **Midterm** (in person on **Oct 14th 2026**, during regular class schedule): 20%
- **Final** (in person on **Dec 16th 2026, 11:20am – 2:10pm**): 35%

- **Bonus**:
  - Note taking: 3%
    - limited and first come first serve. See the head TA's announcement on Piazza.
    - Graded SAT/UNSAT
  - Homework bonus: TBD at each homework

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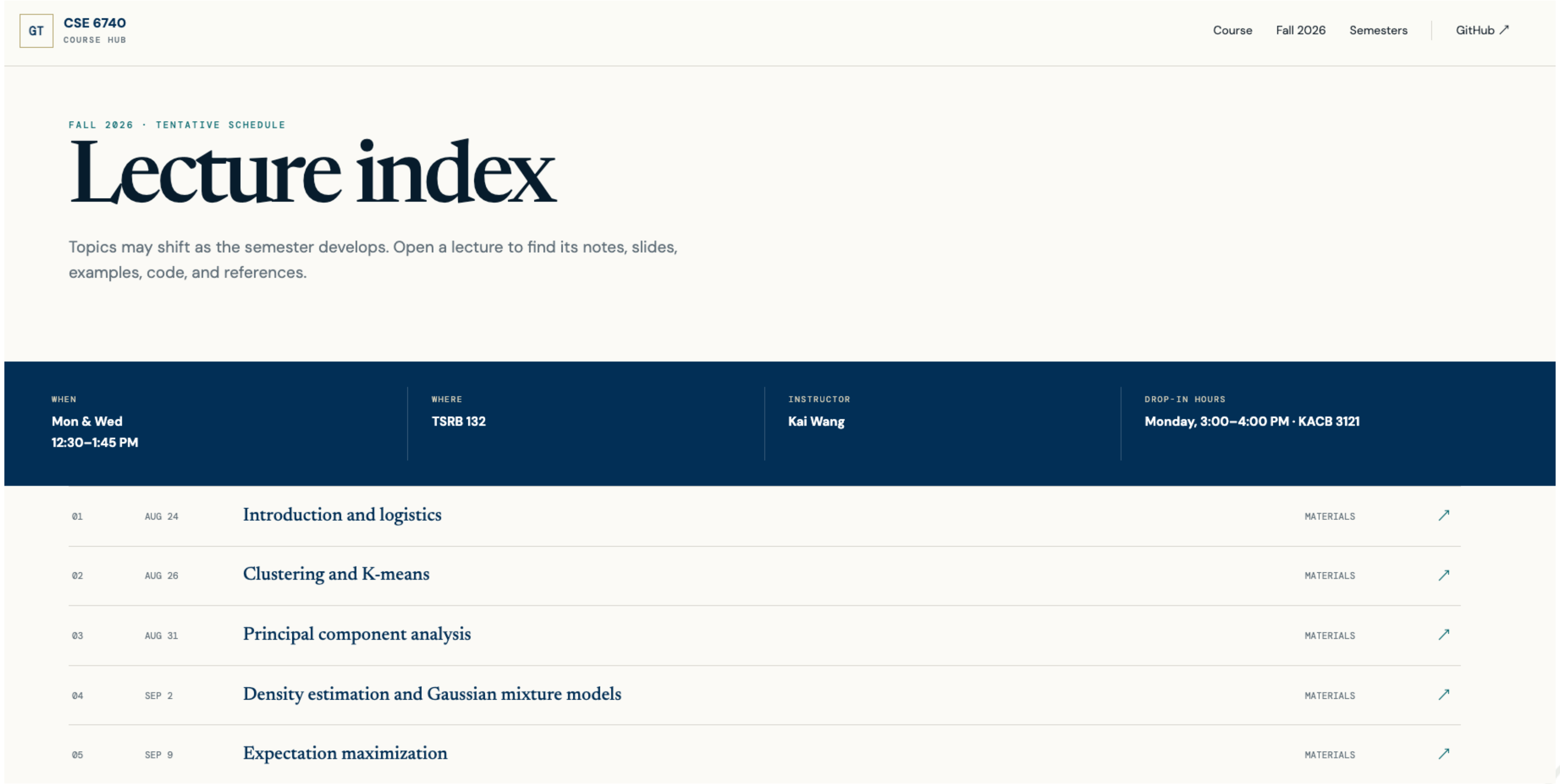
# Note Taking

- https://github.com/GT-KOALA/Gatech-CSE6740



# Final Grade

- Your final grade will be assigned as a letter grade according to the following scale. We will round it to the nearest integer.
  - A 90-100%
  - B 80-89%
  - C 70-79%
  - D 60-69%
  - F 0-59%

# Midterm and Final

- Both will be closed book exams.
- Midterm will be in class on **Oct 14th 2026 (12:30pm– 1:45pm)**
- Final exam: **Dec 16th 2026 (11:20am – 2:10pm)**

- Please plan your schedule earlier!

# Questions